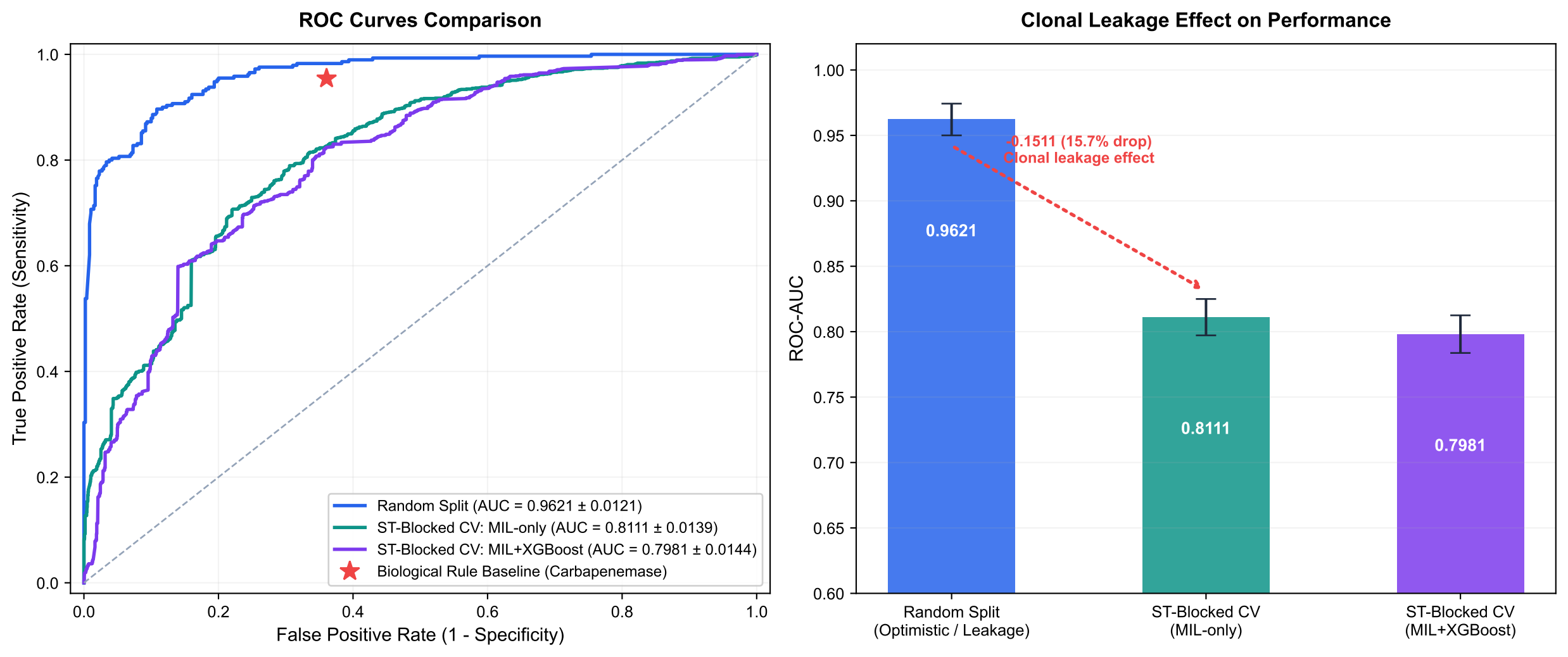
ROC Curves Comparison
True Positive Rate (Sensitivity)
False Positive Rate (1 - Specificity)
0.0
0.2
0.4
0.6
0.8
1.0
Random Split (AUC = 0.9621 ± 0.0121)
ST-Blocked CV: MIL-only (AUC = 0.8111 ± 0.0139)
ST-Blocked CV: MIL+XGBoost (AUC = 0.7981 ± 0.0144)
Biological Rule Baseline (Carbapenemase)
Clonal Leakage Effect on Performance
ROC-AUC
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.9621
0.8111
0.7981
-0.1511 (15.7% drop)
Clonal leakage effect
Random Split
(Optimistic / Leakage)
ST-Blocked CV
(MIL-only)
ST-Blocked CV
(MIL+XGBoost)